

Problem Set 2

Operations Research

Justify your answers and show your calculations. All decision variables are continuous.

Problem 1

A cooperative produces two blends of dried herbs, A and B. Quantities are measured in units of 100 kg, and fractional units are allowed. All production can be sold. The following data apply to one working day:

	Blend A	Blend B	Available
Processing time (hours per 100 kg)	2	1	10 hours
Packaging time (hours per 100 kg)	1	2	8 hours
Selling price (MAD per 100 kg)	900	700	—
Variable cost (MAD per 100 kg)	400	300	—

The cooperative wants to maximize its daily profit. There are no other costs or production limits.

- (a) Define two decision variables and formulate the problem as a linear program in inequality form. Express the objective in hundreds of MAD.
- (b) Solve the problem graphically. Give the vertices of the feasible region and determine whether the optimal solution is unique.
- (c) Write the program in equational form. State the number of variables, the number of equations, and the rank of its constraint matrix.
- (d) Enumerate all basic feasible solutions, giving the corresponding bases and objective values. Use this enumeration to solve the problem.
- (e) Solve the problem by simplex. Show every tableau, identify each entering and leaving variable, and justify the stopping decision.
- (f) Interpret the optimal solution in kilograms, profit in MAD, and unused resources. Explain why the enumeration in part (d) does not list every feasible solution.

Problem 2

A computing centre has a large backlog of three types of data analysis. Processing one GB of each type earns a priority score. Work is divisible, and both the score and resource consumption are proportional to the amount processed.

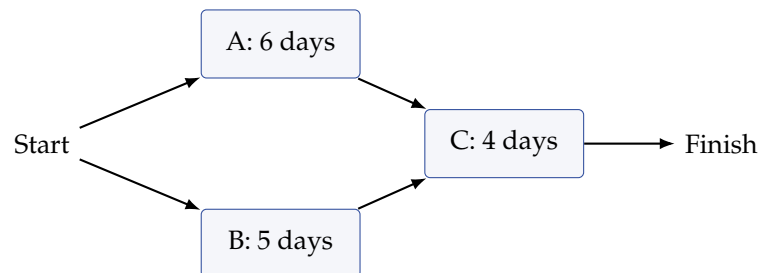
	Type 1	Type 2	Type 3	Daily capacity
CPU time (hours per GB)	1	1	0	4 hours
GPU time (hours per GB)	0	1	1	3 hours
Outbound transfer (GB per GB)	1	0	1	3 GB
Priority score (points per GB)	5	4	5	—

A zero denotes negligible consumption of that resource. There is enough pending work of every type. Setup times and processing order are ignored. The centre wants to maximize its total daily priority score.

- (a) Define the decision variables and formulate the linear program.
- (b) Write the program in equational form and give an initial basic feasible solution and its tableau.
- (c) Solve the program using the simplex method. Show every tableau and justify each pivot.
- (d) Give the optimal quantities and score. Verify the resource use in the original model.
- (e) Prove optimality using the final tableau and identify any unused resources.

Problem 3

A project consists of three activities. A and B can start on day 0 and run in parallel on separate teams. C can start only after both A and B have finished. The project ends when C finishes. There are no other resource conflicts. The normal durations are shown below.



Specialist effort can shorten each activity by any real amount between 0 and 2 days. Shortening A, B, or C costs respectively 4,000, 3,000, or 5,000 MAD per day saved. These costs are linear and additive. The project must finish by the end of day 8, at minimum additional cost.

- Let x_A, x_B, x_C denote the reductions in the three activity durations. Formulate the problem as a linear program, with cost in thousands of MAD.
- Formulate its dual. Specify the domain and the associated primal constraint for each dual variable.
- A manager proposes $(x_A, x_B, x_C) = (2, 1, 0)$. Check feasibility and compute its cost. Exhibit a dual feasible solution proving a lower bound of 8,000 MAD on the minimum cost. What interval do these two solutions give for the minimum cost, and what is the largest possible saving relative to the proposed plan, based only on these bounds?
- Determine optimal primal and dual solutions. Verify complementary slackness and justify optimality.
- Give a schedule meeting the deadline and compute the actual saving relative to the plan in part (c).
- Is the optimal reduction plan unique? Justify your answer.